

ICT702 Data Visualisation

Telling the Truth with Data Visualisation

Module 9



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



Part 2

Biased Data

Biased data occurs when attempting to represent an intended population by drawing data from a non-random sample. There are two types of bias:

- **Selection bias** occurs when data drawn from a non-random sample to represent the intended population.
 - **Example of selection bias:** political polling conducted on landline phone numbers may result in sample biased in terms of age of the respondents
 - **Simpson's paradox** is a type of selection bias. It occurs when a specific trend that appears in subsets of data disappears or reverses when the subsets are aggregated.
- **Survivor bias** occurs when sample data consists of disproportionately large number of observations corresponding to outcomes for a particular event

Example of Simpson's Paradox

Portion of data used to examine relationship between age and income

	A	B	C
1	Age	Income	Location
2	52	\$ 72,600	Dallas
3	58	\$ 53,802	Naples
4	70	\$ 50,405	Naples
5	52	\$ 109,522	San Francisco
6	42	\$ 131,709	San Francisco
7	32	\$ 52,338	Dallas
8	67	\$ 58,456	Naples
9	44	\$ 75,024	Dallas
10	67	\$ 67,999	Naples
11	55	\$ 45,096	Naples
12	42	\$ 54,324	Dallas
13	33	\$ 110,072	San Francisco
14	58	\$ 40,349	Naples
15	65	\$ 41,118	Naples
16	54	\$ 48,012	Naples
17	53	\$ 178,806	San Francisco
18	46	\$ 67,913	Dallas
19	36	\$ 144,141	San Francisco
20	28	\$ 106,496	San Francisco

Study objective: to analyze factors that affect annual incomes in the U.S.

Reported variables: Three variables from three locations

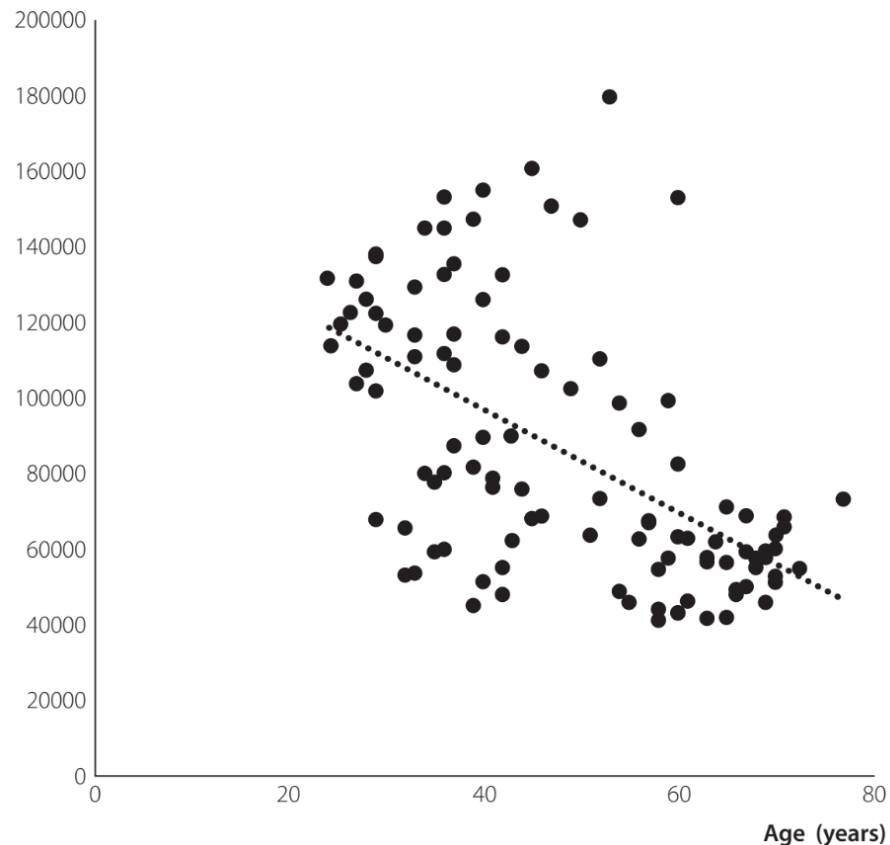
Bias issue: the three locations not necessarily representative of U.S. population

Approach: build a scatter chart of the data aggregated by location and compare it to scatter charts in which the data were disaggregated by location.

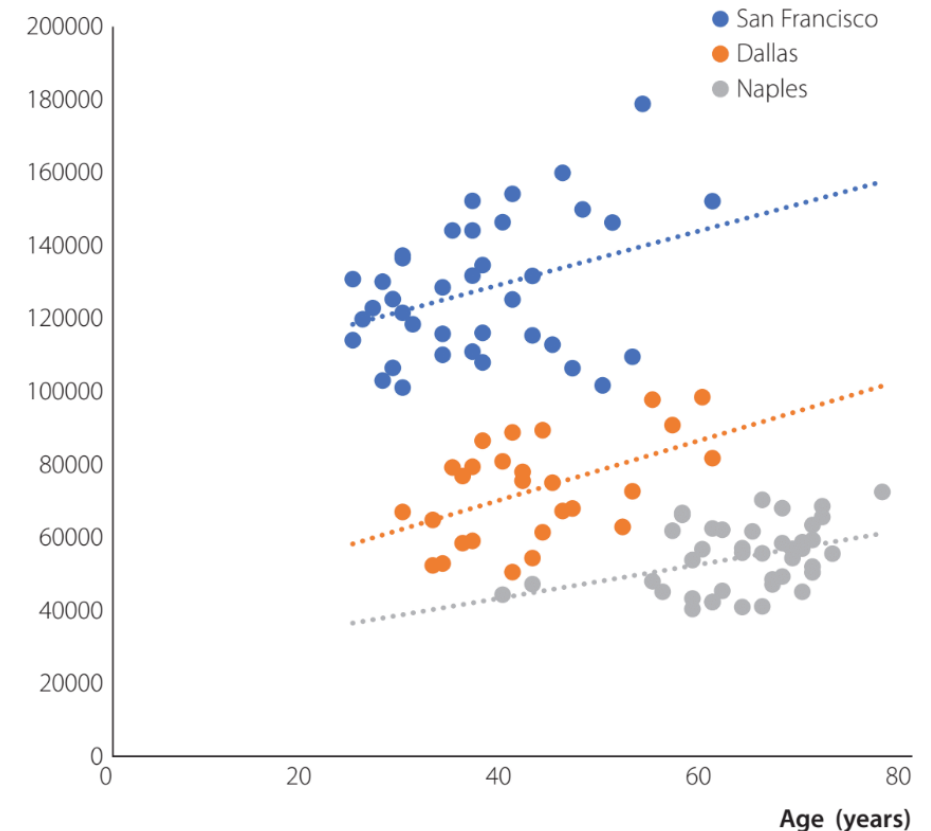
Visualisation of Simpson's Paradox

Scatter charts illustrating how Simpson's paradox may lead to erroneous conclusions

Annual Income (\$)

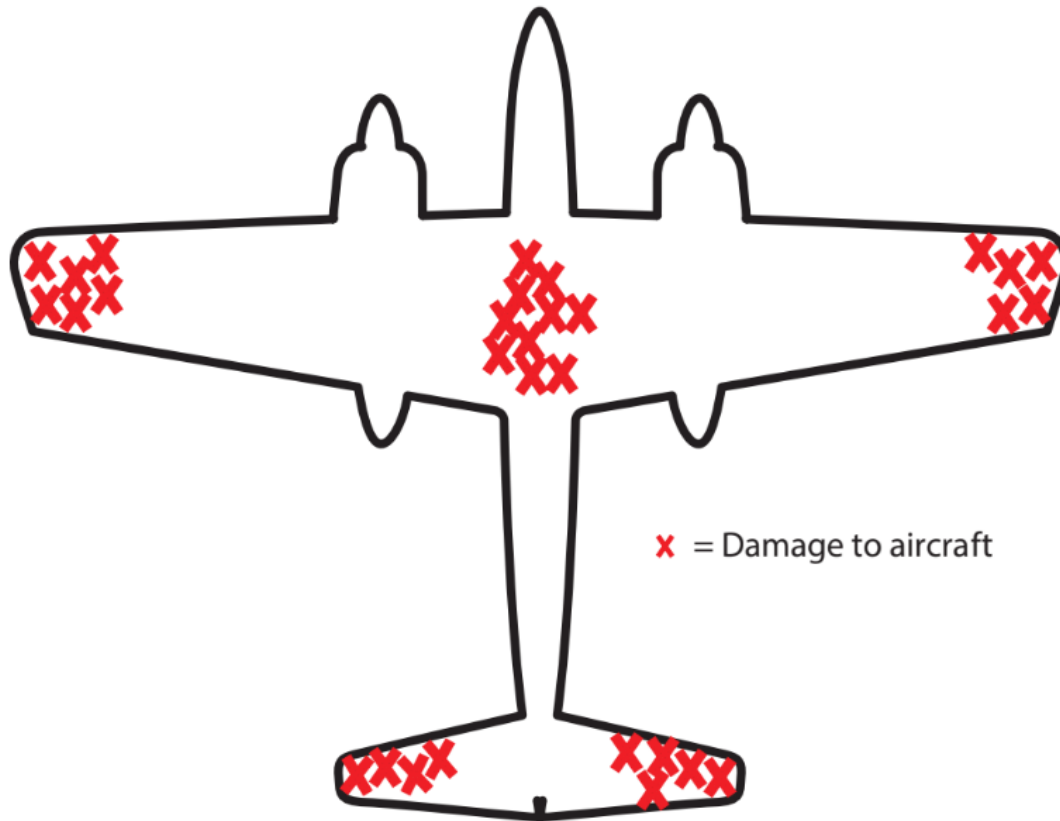


Annual Income (\$)



Origin of Survivor Bias

Depiction of common areas of damage to returning planes in World War II



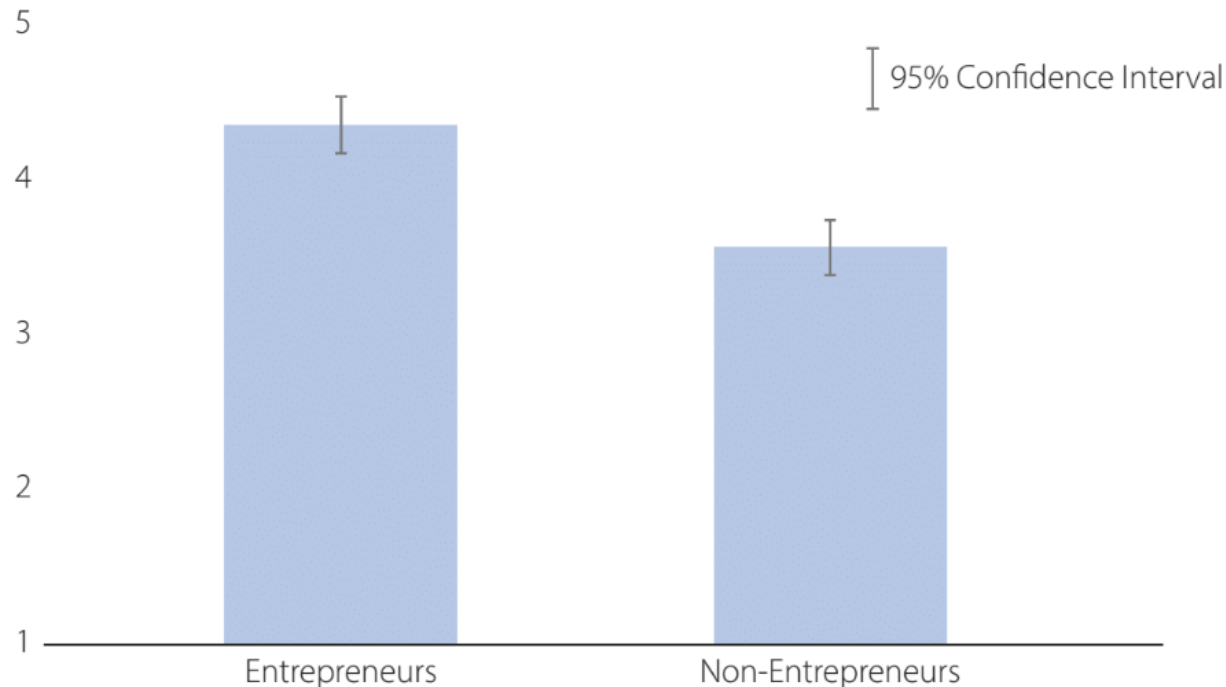
Similar to selection bias, with a survivor bias, the sample data are not representative of the population under study.

Example of Survivor Bias

Comparing average risk tolerance of entrepreneurs to a control group of non-entrepreneurs

Risk Tolerance Comparison for Entrepreneurs and Non-Entrepreneurs

Average Financial Risk Tolerance (1 = Lowest; 5 = Highest)



Research hypothesis:
entrepreneurs are more likely to have a greater risk tolerance than non-entrepreneurs.

Adjusting for Inflation: Example

Portion of data with nominal prices for a gallon of gasoline and price index in US between 1978 and 2017

	A	B	C
1	Year	Nominal Gas Price (per gallon in \$)	Price Index
2	1978	0.65	51.9
3	1979	0.88	70.2
4	1980	1.22	97.5
5	1981	1.35	108.5
6	1982	1.28	102.8
7	1983	1.23	99.4
8	1984	1.20	97.8
9	1985	1.20	98.6
10	1986	0.93	77.0
40	2016	2.20	187.6
41	2017	2.47	211.8

A **price index** is the aggregated price of a basket of products and services.

Inflation refers to the general increase in **nominal prices** over time for the **price index**.

Adjusting for Inflation: Calculations

Adjusting Nominal gasoline prices for Inflation in Excel

Inflation-adjusted prices are calculated for a **base year**.
In this example, 2017 is used:

Infl. adjusted Gas Price 1978

$$= \text{Gas Price 1978} \times \frac{\text{Price Index 2017}}{\text{Price Index 1978}}$$

$$\text{Infl. adjusted Gas Price 1978} = \$0.65 \times \frac{211.8}{51.9} = \$2.66$$

	A	B	C	D
1	Year	Nominal Gas Price (in \$)	Price Index	Inflation Adjusted Price (in 2017 \$)
2	1978	0.652	51.9	=B2*(\$C\$41/C2)
3	1979	0.882	70.2	=B3*(\$C\$41/C3)
4	1980	1.221	97.5	=B4*(\$C\$41/C4)
5	1981	1.353	108.5	=B5*(\$C\$41/C5)
6	1982	1.281	102.8	=B6*(\$C\$41/C6)
7	1983	1.225	99.4	=B7*(\$C\$41/C7)
8	1984	1.198	97.8	=B8*(\$C\$41/C8)
9	1985	1.196	98.6	=B9*(\$C\$41/C9)
10	1986	0.931	77	=B10*(\$C\$41/C10)
40	2016	2.204	187.602	=B40*(\$C\$41/C40)
41	2017	2.469	211.77	=B41*(\$C\$41/C41)

	A	B	C	D
1	Year	Nominal Gas Price (in \$)	Price Index	Inflation Adjusted Price (in 2017 \$)
2	1978	0.65	51.9	2.66
3	1979	0.88	70.2	2.66
4	1980	1.22	97.5	2.65
5	1981	1.35	108.5	2.64
6	1982	1.28	102.8	2.64
7	1983	1.23	99.4	2.61
8	1984	1.20	97.8	2.59
9	1985	1.20	98.6	2.57
10	1986	0.93	77.0	2.56
40	2016	2.20	187.6	2.49
41	2017	2.47	211.8	2.47

In cell D2 of the Excel spreadsheet, the above equation is implemented as:

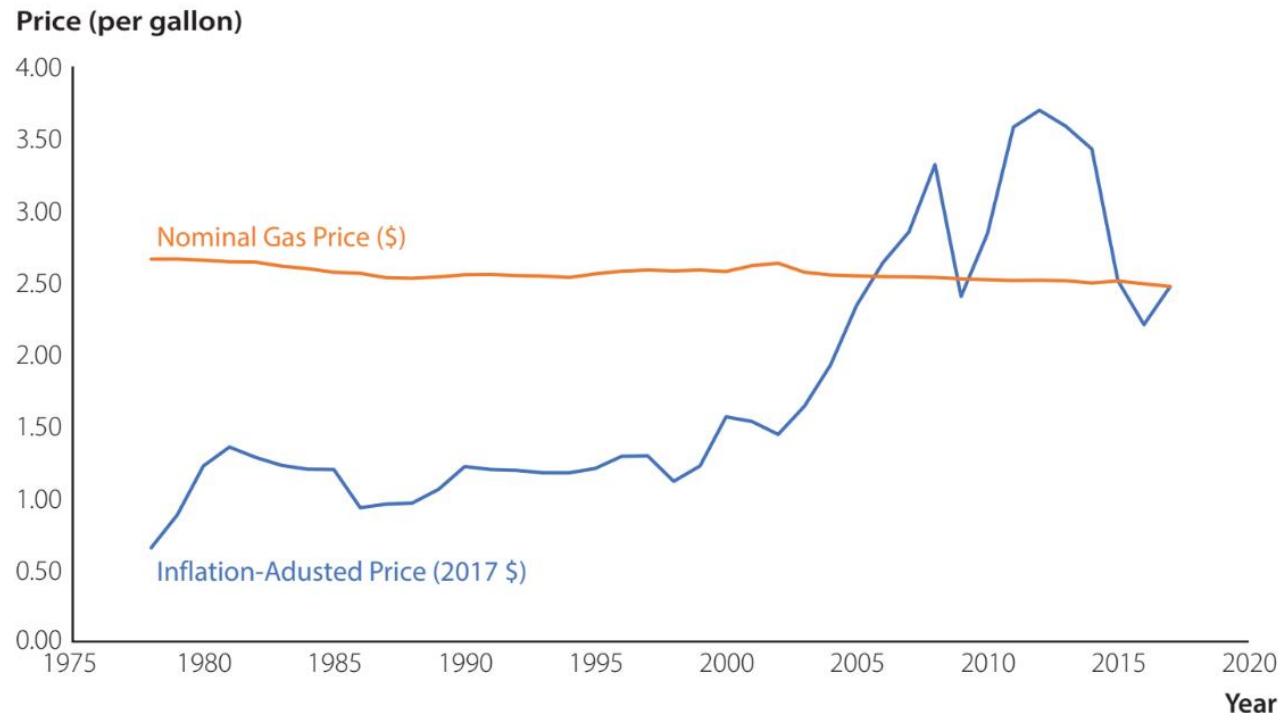
$$=B2*(C41/C2)$$

And pasted it to cells D3:D41.

Adjusting for Inflation: Visualisation

The inflation-adjusted price per gallon of gasoline in the US has decreased between 1978 and 2017

Comparing Nominal and Inflation—Adjusted Gas Prices between 1978 and 2017



Nominal values can be adjusted to account for inflation using any year in the time series.

The selection of the base year generally depends on the comparisons to be made.

End of Part 2

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